
1. Install SonarQube (fresh setup)

```
sudo apt update
sudo apt install unzip wget openjdk-17-jdk -y

cd /opt
sudo wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.5.1.90531.zip
sudo unzip sonarqube-*.zip
sudo mv sonarqube-* sonarqube
```

Create user

```
sudo useradd -r -s /bin/bash sonar
sudo chown -R sonar:sonar /opt/sonarqube
```

2. Configure SonarQube service

```
sudo vi /etc/systemd/system/sonarqube.service
```

Paste:

```
[Unit]
Description=SonarQube service
After=network.target

[Service]
Type=forking
ExecStart=/opt/sonarqube/bin/linux-x86-64/sonar.sh start
ExecStop=/opt/sonarqube/bin/linux-x86-64/sonar.sh stop
User=sonar
Group=sonar
Restart=always
LimitNOFILE=65536
LimitNPROC=4096
PIDFile=/opt/sonarqube/temp/sonar.pid
```

```
[Install]
```

WantedBy=multi-user.target

3. Increase system limits (IMPORTANT)

`sudo vi /etc/sysctl.conf`

Add:

```
vm.max_map_count=262144
fs.file-max=65536
```

Apply:

`sudo sysctl -p`

4. Start SonarQube

```
sudo systemctl daemon-reload
sudo systemctl start sonarqube
sudo systemctl enable sonarqube
```

Check:

`systemctl status sonarqube`

5. Access UI

Open:

`http://<YOUR-IP>:9000`

Login:

admin / admin

- Create a **Project**
 - Generate a **Token**
-

6. Install Sonar Scanner

```
cd /opt
sudo wget
https://binaries.sonarsource.com/Distribution/sonar-scanner-cli/sonar-scanner-cli-5.0.1.3006-linu
x.zip
sudo unzip sonar-scanner-cli-*.zip
sudo mv sonar-scanner-* sonar-scanner
```

Add to PATH:

```
echo 'export PATH=$PATH:/opt/sonar-scanner/bin' >> ~/.bashrc
source ~/.bashrc
```

Check:

```
sonar-scanner -v
```

7. Clone your NodeJS / NextJS app

```
cd /var/www
git clone https://github.com/akkimahesh/nextjs-application.git
cd nextjs-application
npm install
```

8. Create sonar-project.properties

vi sonar-project.properties

Paste:

```
sonar.projectKey=nextjs-app  
sonar.projectName=NextJS App  
sonar.projectVersion=1.0
```

```
sonar.sources=.
```

```
sonar.sourceEncoding=UTF-8
```

```
sonar.host.url=http://<YOUR-IP>:9000  
sonar.token=YOUR_TOKEN
```

```
sonar.exclusions=node_modules/**,.next/**,out/**,coverage/**
```

9. Run Scan

sonar-scanner

10. View Results

Go to:

<http://<YOUR-IP>:9000>

 Open your project

 See:

- Bugs 
- Vulnerabilities 
- Code smells 
- Coverage 



Common Issues (VERY IMPORTANT)

Connection refused

👉 Fix:

```
sudo systemctl status sonarqube
```

Not authorized

👉 Fix:

- Use correct token
 - OR create project manually in UI
-

Port not reachable

👉 Fix AWS security group:

Port 9000 → Allow Anywhere (0.0.0.0/0)

Scanner not found

👉 Fix PATH:

```
export PATH=$PATH:/opt/sonar-scanner/bin
```



Final Flow (Quick Summary)

1. Install SonarQube
2. Start service
3. Open UI → create project + token

4. Install scanner
 5. Add `sonar-project.properties`
 6. Run `sonar-scanner`
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If you want next step, I can help you with:

- ✓ Jenkins pipeline integration
- ✓ GitHub Actions auto scan
- ✓ Docker setup for SonarQube

Just tell 👍